MILITARY SPECIFICATION
CASTINGS; MAGNESIUM INOCULATED, MALLEABLE IRON FOR ARTILLERY, WARHEAD AND MORTAR AMMUNITION COMPONENTS

1. SCOPE

1.1 Scope.- This specification covers magnesium inoculated malleable iron castings for artillery and mortar ammunition components.

1.2 Classification.- The castings shall be manufactured from the grades as specified in table I and shall conform to the chemical composition requirements of 3.5.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-P-10025	- Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification For

STANDARDS

Federal

FED-STD-151	- Metal; Test Methods

Military

MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Melting process.- The iron shall be made by one of the following processes: air furnace, open hearth, electric furnace or any of the multiplex processes.

3.1.1 The iron shall be inoculated with unalloyed magnesium to comply with 3.5 and shall be cast in the metastable white iron condition.

3.2 Molding.- Molds and cores shall be made on semi-automatic or automatic molding equipment, approved by the procuring activity to assure that molds and cores of consistent properties will be obtained on a reproducible basis.

3.3 Heat-treating.

3.3.1 Annealing (graphitizing).- Castings shall be annealed in a controlled atmosphere furnace approved by the procuring activity.

3.3.2 Quenching and tempering.- When castings are hardened by austenitizing, quenching and tempering, liquid quenching shall be employed, followed by a suitable tempering procedure to obtain the specified mechanical properties.

3.3.3 Optional heat treatment.- Grade 60 material may be obtained with a suitable annealing heat treatment with manganese alloyed malleable iron. This heat treatment shall be performed on the whole casting in such a manner as to result in uniform mechanical properties throughout the individual part and throughout the heat-treat lot.

3.3.4 Subsequent heating.- After the final heat treatment to obtain the specified mechanical properties, castings shall not be heated to a temperature sufficient to alter these properties, except in localized areas for brazing.

3.4 Mechanical properties.- The castings shall meet the mechanical property requirements shown in table I.

Table I

MECHANICAL PROPERTY REQUIREMENTS

Grade	Yield Strength (psi) Min (0.1% offset)	Tensile Strength (psi) Min	Elongation Min %	Brinell Hardness*
50	48,000	65,000	18	135-175
60	60,000	80,000	5	187-241
80	80,000	170,000	3	235-269

*Brinell hardness range is advisory only.

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3.5 Chemical composition.- Magnesium inoculated malleable iron shall conform to one of the compositions contained in table II.

Table II

CHEMICAL COMPOSITION REQUIREMENT

Element	Grades 50, 80 (Percent)	Grades 60, 80 (Percent)
Carbon	2.30 - 2.50	2.30 - 2.50
Silicon	2.50 - 2.75	1.70 - 2.00
Manganese	0.40 Max	0.90 - 1.25
Magnesium	0.025 Min	0.025 Min
Sulfur	0.02 Max	0.02 Max
Phosphorous	0.08 Max	0.08 Max

3.6 Foundry control.- Castings shall be produced under foundry control approved by the procuring activity. Foundry control shall consist of examination of castings by radiographic, micrographic, macrographic, ultrasonic or other approved means until the pattern and mold design, and melting, molding and pouring practices have been established to produce castings meeting the quality standards furnished or approved by the procuring activity. Foundry control shall also consist of those process controls necessary to assure that castings of the required quality will be consistently obtained. When foundry practice and process controls have been so established, production practies shall not be changed without demonstrating to the satisfaction of the procuring activity that the changes do not adversely affect the quality of the casting.

3.7 Soundness.- The soundness of castings shall meet standards furnished or approved by the procuring activity using inspection methods specified in the item specification and by the procuring activity. The quality standard shall consist of a sample casting or castings, radiographs, photomicrographs, macrographs, other standards specified by the procuring activity or a combination of these which shall clearly indicate the quality to be obtained.

3.8 Repair of castings.- Defective castings shall not be plugged or welded.

3.9 Surface finish and condition.

3.9.1 Machined surfaces.- Machined surfaces shall meet the surface finish requirements specified on the applicable drawing.

3.9.2 As-cast surfaces.- As-cast surfaces on the finished part shall be as specified on the applicable drawing and specification for the end item and shall be free of scale resulting from the casting process and subsequent heat-treatment operations. Surface defects such as pits, sandholes, porosity, core line impressions or any similar defects

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greater than those permitted by the standards which are provided or stipulated by the procuring activity shall be cause for rejection of the casting. Grinding to remove surface defects shall be permitted to minimum wall thickness requirements provided that thickness measurements are taken in each ground area to insure compliance with drawing requirements. Grinding shall be done in such a manner as to provide a smooth contour and shall have no sharp corners or edges.

3.10 Decarburization.- Unless otherwise specified by the procuring activity, the depth of decarburization and affected microstructure measured from any cast surface shall not be greater than .035 inch, when tested in accordance with paragraph 4.3.4. The decarburization requirement does not apply to Grade 60 material with a ferritic matrix.

3.11 Mottle.- Castings shall be free of mottle (primary graphite) when tested in accordance with 4.3.3.

3.12 Dimensional tolerance.- Dimensions and dimensional tolerances of finished castings shall be as specified in the applicable drawing.

3.13 Identification markings.- The identification mark of the manufacturer, pattern number, date and time (shift) of casting shall be cast into the surface of each casting using a format and location as approved by the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection provisions.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 Inspection provisions.

4.2.1 Casting lot.- A casting lot shall consist of castings poured from the same inoculated ladle.

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4.2.1.1 Heat treatment lot.- A heat treatment lot may consist of any number of casting lots not to exceed four hours production of the final heat treat furnace. All castings shall be heat treated at the same temperatures for the same lengths of time or at the same conveyor speed. The lot thus constituted shall be the basis for selection of sample castings to be used for the tension tests prescribed by 4.3.1.2.

4.3 Inspection testing.

4.3.1 Mechanical properties.

4.3.1.1 Hardness.- In order to insure uniformity of heat treatment and to establish a hardness range to assure compliance with specified mechanical properties, each casting, after final heat treatment, shall be subjected to a hardness test with a Brinnell hardness tester at a location on the casting specified or approved by the procuring agency.

4.3.1.1.1 Alternate hardness testing.- A direct reading Brinell hardness testing machine may be substituted for the Brinelloscope method, provided the same diameter ball is used and the same load is applied as is used in standard Brinell testing, and provided that at least 2,000 castings are tested by both methods to establish a correlation between the two methods. After a correlation has been established, the diameter of the indentation in every tenth casting shall be measured with the Brinelloscope to assure continued correlation. Other methods of hardness testing not employing indentors may be employed with the approval of the procuring activity.

4.3.1.2 Tension test.- Two castings from each lot representing the hardest and softest castings as determined by the hardness test outlined in 4.3.1.1 or 4.3.1.1.1, shall be subjected to the tension test. Two specimens from each casting shall be prepared as specified in 4.4.1.2 and tested. If all specimens comply with the requirements of 3.4, the hardness range for the lot will have been established. Failure of one or more tension test specimens to comply with the requirements shall be cause for rejection of the lot represented by the specimen.

4.3.1.3 Retest.- When one or more tension test specimens fail to comply with the requirements of 3.4 and do not qualify for retest under 4.4.1.2.2, a new hardness range shall be established. The rejected lot may, at the discretion of the contractor, be reheat treated prior to establishing a new hardness range. The tension test, in compliance with 4.3.1.2 shall be conducted to determine the acceptability of the new hardness range. When the hardness range has been established, all castings whose values fall outside this range shall be rejected.

4.3.2 Soundness.- Each projectile body casting or any other casting that is subject to being loaded with high explosives or chemical filler, shall be tested in accordance with 4.4.2 or as prescribed by the procuring activity. Failure of any casting to comply with 3.7 shall be cause for rejection of the individual casting. When inspection is performed on

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a sampling basis the lot shall be rejected if the rejection criteria has been reached or exceeded.

4.3.3 Mottle.- A minimum of two castings or test coupons of a section size equal to or greater than the section size of the projectile casting, selected for every 30 minutes or fraction thereof of continuous pouring time, shall be tested in accordance with 4.4.3. Failure of any samples to comply with 3.11 shall be cause for rejection of all castings poured since the last acceptable samples were taken.

4.3.4 Decarburization.- One casting from each heat treat lot shall be tested in accordance with 4.4.4. Failure of the sample to comply with 3.10 shall be cause for rejection of the lot. Lots of castings that have complied with the requirements of 3.10 and are subsequently reheat treated shall be re-examined as prescribed above. One of the castings selected for testing of mechanical properties may be used for this purpose.

4.3.5 Identification marking.- At least one casting from each pour shall be visually examined for legibility of markings and for proper location to determine compliance with 3.13.

4.3.6 Chemical analysis.- Each casting lot shall be subjected to a chemical analysis for compliance with 3.5. Failure of a casting lot to comply with 3.5 will be cause for rejection of the casting lot.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Hardness.- The Brinell hardness test shall be performed in accordance with procedures prescribed in FED-STD-151. Due to the nature of the microstructure of pearlitic malleable iron, indentor balls of less than 10 millimeter diameter shall not be used. Alternate hardness test methods shall be performed by procedures established or approved by the procuring activity.

4.4.1.2 Tension test.- The tension test specimen shall be the largest obtainable standard round specimen prescribed by FED-STD-151 (types R1 thru R5) which the size of the casting and location of the specimen will permit. If the size or geometry of the casting does not permit a standard round specimen, a specimen as specified or approved by the procuring activity shall be used.

4.4.1.2.1 Yield strength, tensile strength and elongation.- Yield strength shall be determined by the offset method as defined in FED-STD-151. The offset shall be 0.1% (0.001 inch per inch of gage length). Tensile strength and elongation on each specimen shall be determined by utilizing the criteria for these tests prescribed by FED-STD-151.

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4.4.1.2.2 Replacement test specimens.- If a tension test specimen fails to meet the requirements of 3.4 due to an identifiable defect in the tensile specimen which is acceptable under provisions of 3.7 and 4.3.2 or due to a fracture occurring under those conditions allowing for retests for elongation in FED-STD-151, another specimen shall be machined from the same casting and tested. In the instance of failure due to an acceptable metal defect, only one replacement specimen shall be permitted and failure of the replacement specimen to comply with the requirements of 3.4 shall be cause for rejection of the represented lot. Results of all tests, original and replacement tests, shall be recorded and specimens which failed to meet the requirements due to the above conditions shall be identified and retained by the contractor and made available to the procuring activity for examination upon request.

4.4.2 Soundness.- For those applications where 100% inspection is required, castings shall be inspected by ultrasonics. If the geometry of the casting does not permit the use of ultrasonics, some form of radiography, magnetic particle inspection or some other form of inspection shall be specified and approved by the procuring activity. Inspection methods for less critical components shall be as specified by the procuring agency.

4.4.3 Mottle.- The samples selected in accordance with 4.3.3 shall be broken in such a manner that the fracture will extend through the heaviest section. The fracture shall be visually examined for compliance with 3.11.

4.4.4 Decarburization.- From the samples selected in accordance with 4.3.4 a suitable specimen shall be taken and prepared for study of the microstructure. The depth of decarburization shall be measured by viewing the specimen at 100 magnifications. The zone of affected microstructure shall include any structure which is essentially different from the matrix structure.

4.4.5 Chemical analysis.- Chemical analysis for compliance with 3.5 shall be performed by wet analysis or spectrometry or by any other method approved by the procuring activity.

5. PREPARATION FOR SHIPMENT

5.1 Packing.

5.1.1 Level A.- Not applicable.

5.1.2 Level B.- No applicable.

5.1.3 Level C.- Castings shall be packed in accordance with the requirements of MIL-P-10025.

5.2 Labeling and marking.- Labeling and marking shall be in accordance with the requirements of MIL-P-10025.

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5.3 Data cards.- When specified by the contracting officer data cards shall be required as specified in MIL-STD-1167.

6. NOTES

6.1 Intended use.- Magnesium inoculated malleable iron castings purchased to this specification are intended for use in artillery, warhead and mortar ammunition components.

6.2 Tension test specimens.- Separately cast test coupons for tension test specimens are not always representative of the mechanical properties present in the casting and therefore, shall not be used in tension tests for determining compliance with mechanical property requirements unless specifically authorized by the procuring activity.

6.3 Government owned tooling.- Due to the technical aspects involved, the mold patterns and core boxes shall be Government owned tooling.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1395-A027

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